

CP-even kinetic freedom at fixed quark masses

Nine polynomial covariants, positive metrics and exact
heavy-mediator matching

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A constructive operator-space theorem with fixed-spectrum countermetrics, a shared
non-Abelian source label and a stable heavy-family branch.

Result and assumptions

A complete Hermitian quark metric can be generated by nine CP-compatible polynomial covariants with real coefficients. When their CP triangle is nonzero, they span all nine real metric directions at the reference vacuum. Small positive changes of the bare right-handed metric can preserve a previously solved scalar vacuum, its superpotential and all six quark masses while changing every physical CKM parameter. This is an algebraic statement about allowed operators, rather than a claim that quantum corrections have a particular size. It makes a protecting model responsible for kinetic matching as well as scalar and Yukawa interactions.

The assumptions are linear unitary internal flavor transformations, flavor-covariant functions X and Z with the same left flavor representation, a full-rank X with distinct singular values, a nonzero CP triangle, and independent real coefficients for the complete allowed matter kinetic operator space. X and Z may have different right representations and ordinary phase charges: those cancel from $R=X^\dagger X$ and $S=X^\dagger Z Z^\dagger X$. The degree bound below requires polynomial covariants. Matched analytic Yukawa functions obey the spanning argument but require an effective-field-theory expansion before a polynomial degree can be assigned. Degenerate spectra, CP-conserving vacua, coefficient restrictions imposed by a UV theory and additional restrictions beyond these internal transformation laws are outside this theorem.

The numerical demonstration uses the already minimized nonorthogonal joint vacuum from commit f76767e. The golden-angle test is a separately initialized attack fixture. It does not turn the nominated angle or golden coefficient into a dynamical prediction. The accompanying common-label construction supplies a justified interaction and a stable heavy-family branch, but has not been solved as a complete joint three-family model.

Nine real polynomial covariants

$$R = X^\dagger X, \quad S = X^\dagger Z Z^\dagger X, \quad \det \mathcal{M} = \tau^3$$

Let $R=X^\dagger X$ and $S=X^\dagger Z Z^\dagger X$. Both transform by conjugation on the right flavor space of X . Under generalized CP they transform into their complex conjugates conjugated by the corresponding unitary CP matrix. Fix the three positive distinct eigenvalues $r_1 < r_2 < r_3$ of R at a reference vacuum as real numerical constants. Define $P_i(R)$ as the product over j unequal to i of $(R - r_j I)/(r_i - r_j)$. The P_i are genuine matrix polynomials with fixed coefficients, rather than eigenprojectors recomputed as the fields vary. At the reference vacuum they are the three orthogonal rank-one spectral projectors.

Take the three P_i and, for each pair $i < j$, the two Hermitian covariants $B_{ij}=P_i S P_j + P_j S P_i$ and $D_{ij}=P_i S^2 P_j + P_j S^2 P_i$. Every coefficient is real, every expression is Hermitian, and each expression has the required family and CP transformation law. In particular, there is no explicitly imaginary coefficient inserted to manufacture a missing phase direction.

In an R eigenbasis put $\tau = \text{Im}(S_{12} S_{23} S_{31})$. For the pairs (12),(13),(23), the real-coordinate determinants of the two complex off-diagonal entries S_{ij} and $(S^2)_{ij}$ are respectively $-\tau, +\tau, -\tau$. The diagonal entries of S only add real multiples of S_{ij} and cancel from these determinants. The three diagonal projectors span the diagonal subspace. Consequently the nine-generator coordinate determinant is τ^3 in coordinates consisting of the three diagonals followed by real and imaginary off-diagonal entries. It is $8\tau^3$ if both off-diagonal coordinates are multiplied by square-root two to obtain Hilbert-Schmidt orthonormal coordinates. This proves completeness when τ is nonzero. The exact symbolic identities are independently checked in the driver and focused tests.

An invariant certificate of physical CP

$$\text{Im Tr}(R^2 S^2 R S) = \Delta(R) \tau$$

The reference-basis condition can be written without choosing an eigenbasis: $\text{Im Tr}(R\text{-squared } S\text{-squared } R S) = \Delta(R) \tau$, where $\Delta(R) = (r_2 - r_1)(r_3 - r_1)(r_3 - r_2)$. This identity is checked by symbolic expansion of a general Hermitian S . A nonzero left-hand side certifies that the covariants do not admit a simultaneous CP-real frame.

If X and Z are the canonically normalized up and down Yukawa functions, S in the R eigenbasis is the up singular-value diagonal matrix times the down left mass-squared matrix times the same diagonal matrix. Its triangle imaginary part is the product of the three squared up singular values times the triangle imaginary part of the down matrix in the up mass basis. For nondegenerate spectra that latter quantity is proportional to the physical Jarlskog invariant and the down spectral Vandermonde. Thus nonzero physical weak CP supplies the required spanning condition for the matched Yukawa functions. For the finite polynomial covariants used in the explicit heavy-mediator example, the condition is checked directly; it is not inferred merely from the existence of a phase in their entries.

Positive completion and finite degree

$$B(\Phi) = (I + T(\Phi))^2 + \eta I \geq \eta I$$

At the reference vacuum orthonormalize the three off-diagonal pairs using real Hilbert-Schmidt inner products. This uses fixed real normalization constants. Together with the three projectors the resulting Q_a form a real orthonormal Hermitian basis. Given any positive target right-handed metric B_0 , choose a positive η strictly smaller than its least eigenvalue, set $T_0 = \text{square-root}(B_0 - \eta I) - I$, and expand T_0 as the sum of the nine Q_a times the real coefficients $\text{Tr}(Q_a T_0)$. Use the same coefficients and fixed polynomial recipes at every field value to define $T(\phi)$.

The metric $B(\phi) = (I + T(\phi))^2 + \eta I$ equals B_0 at the reference point and is bounded below by ηI at every field value. Positivity follows because T is Hermitian. This is a positive completion of the right-handed block. It is not a global positivity theorem for the entire Kahler metric, whose mixed matter-flavon blocks appear away from zero matter. Near zero matter backgrounds those mixed blocks vanish and positivity of the complete metric follows locally by continuity.

If X has flavor-field degree p and Z degree q , then R has degree $2p$, S has degree $2p+2q$, and each interpolating P_i has degree at most $4p$. The highest-degree generator $P_i S\text{-squared } P_j$ plus its adjoint has degree at most $12p+4q$. Squaring the Hermitian completion gives an upper bound $24p+8q$ on the flavor-field degree of B . A matter bilinear adds two matter fields to the operator degree. The explicit down-sector example uses $X = L C_d$ and $Z = C_u$, hence $p=2, q=1$ and a maximum flavor-field degree of 56. High-degree coefficients and reference spectral denominators can be large; no naturalness or loop-size estimate follows from this existence proof.

Fixed masses with full heavy matching

$$B = Y_0^\dagger H_{\text{target}}^{-1} Y_0 - A^\dagger A, \quad Y Y^\dagger = H_{\text{target}}$$

The bare light right-handed quarks mix with heavy quarks through a fixed holomorphic null frame $N=(I,-A)$. Let Y_0 be the corresponding unnormalized Yukawa matrix. With the old identity bare metric the effective metric is $K_0=I+A\dagger A$ and the matched Yukawa is $Y_0 K_0$ to the power $-1/2$. Replace only the bare light-quark metric by B . The effective null-frame metric becomes $K=B+A\dagger A$.

For any sufficiently nearby positive target left mass-squared matrix H_{target} , choose $B=Y_0\dagger H_{\text{target}}$ to the power -1 $Y_0 A\dagger A$. Then $K=Y_0\dagger H_{\text{target}}$ to the power -1 Y_0 and, for invertible Y_0 , the canonically matched Yukawa obeys $Y Y\dagger=H_{\text{target}}$ exactly. At the reference target this B is the identity, so positivity persists for a neighborhood of targets. Taking $H_{\text{target}}=U H_{\text{down}} U\dagger$ preserves all down masses. The up sector is untouched. Relative unitary rotations cover the four physical mixing directions after rephasings, at a generic three-family point.

Every such B can be realized by the CP-compatible positive polynomial completion above. In the chiral-superfield convention its transposed right-handed metric is used if required by the definition of the physical right-handed column; the positive spectrum and covariance statement are unchanged. This matching is performed before eliminating the heavy fields. It does not impose an arbitrary matrix after canonical normalization and call that matrix a symmetry-derived contraction.

At zero quark, heavy-quark and Higgs backgrounds, the additional Kahler term is a matter bilinear and vanishes. Its mixed derivatives vanish there as well, and its flavon block does not change. Therefore the existing scalar stationary point and flavor Hessian remain unchanged at this tree-level background. The claim does not include an electroweak vacuum calculation or a radiatively corrected scalar vacuum.

Constructive checks on the solved joint vacuum

The driver reconstructs the unchanged up and down superpotential matrices and the complete down heavy null frame of the prior joint solution. It uses the polynomial covariants $X=L C_d$ and $Z=C_u$ on the bare down-quark label space. Six anti-Hermitian rotations in the up mass basis, each evaluated with both signs at magnitude 0.00001, produce twelve positive bare metrics. All twelve are realized with real polynomial coefficients and matched again through their heavy null-frame metrics.

The four singular values of the fractional-observable, column-normalized response matrix are 1.94061, 1.18728, 0.736859, 0.530491. These normalization-dependent numbers certify four independent observable directions; they are not physical sensitivity eigenvalues. The greatest logarithmic residual among all six quark singular values is below 7 times 10 to the power -13 in double-precision SVD. The nine-basis orthonormality error and positive metric reconstruction errors are below 10 to the power -75 in 90-digit arithmetic. After explicitly Hermitianizing the floating-point target mass matrix, the exact-matching residual is likewise checked at multiprecision. The driver records each metric, its eigenvalue bound, its nine real coefficients and all resulting observables.

A separate golden-chart fixture starts with $C=(3\text{-square-root five})/2$, physical phase 66 degrees, $|V_{us}|=0.2253$ and $|V_{cb}|=0.0409$. Holding both anchors, the phase and all six masses fixed, two CP-compatible positive metrics change C by -0.1% and $+0.1\%$. Their operator-norm distance from the identity is about 0.00343545, and their least eigenvalue exceeds 0.99657. This is a concrete allowed deformation of an initialized exact relation. It neither predicts these inputs nor establishes the typical size of kinetic corrections in a natural UV completion.

A shared non-Abelian source label

$$Z_8 = Z - \frac{\text{Tr} Z}{3} I, \quad V_{\text{eff}} = -\frac{k^2}{M^2} \|Z_8\|^2$$

The positive interaction candidate replaces three independently charged source columns in each sector by a single matrix source with one common phase charge. Introduce a common label group $H=3.A_6$ and sources $A=(3_u, 3_{\bar{H}})$, $B=(3_d, 3_{\bar{H}})$, together with one right-handed H triplet per quark charge sector. The common group acts on source labels as a representation, rather than permitting three arbitrary column coefficients. The exact singlet multiplicity in triplet times conjugate triplet is one, so the corresponding quark-source contraction has one coupling per sector.

For the link $L=(3_u, 3_{\bar{d}})$, $Z=A^\dagger L B$ transforms by conjugation under H . The endomorphism representation decomposes as $1+8$. The two quadratic dressed channels are the singlet norm $|\text{Tr} Z|^2$ and the traceless norm $\text{Tr}(Z^\dagger Z) - |\text{Tr} Z|^2/3$. Exact averaging over all 1080 group elements gives singlet multiplicities 1 in triplet times conjugate triplet, 1 in adjoint squared and 2 in adjoint cubed. These are representation-derived coupling counts.

A complex adjoint scalar S_H , with the phase charge of Z , permits M -squared $\|S_H\|^2 - 2k \text{Re} \text{Tr}(S_H^\dagger Z_8)$, where $Z_8 = Z - (\text{Tr} Z)I/3$. Eliminating S_H at this leading order gives $-(k^2/M^2)\|Z_8\|^2$. This correlates the two norm coefficients through the fixed $-1/3$ trace-subtraction ratio. The ratio is a Clebsch coefficient and is not the golden coefficient. The source-link-mediator term is quartic in elementary scalar fields. Additional allowed mediator quartics and mediator norm couplings to the source and link bilinears must be included in a complete UV matching; the leading exchange is not a license to omit them.

A stable heavy-family branch

The same sextic invariant tensor gives a matrix-source invariant $I_6(A)$. Include the second allowed holomorphic sextic through $W_A = I_6(A) + c \det(A)^2$, with real c . In dimensionless units consider $V_A = \|\text{gradient } W_A\|^2 - (180+6t)\|A\|^2 + 2t \text{Re } W_A$. The rank-one configuration $A=\text{diag}(1,0,0)$ is stationary, as are its finite group rotations. The determinant begins at second order around rank one, so its square and its first three derivatives vanish there. Hence its coefficient c does not alter the local Hessian.

In canonically normalized real coordinates the eighteen mass coefficients are: radial $1440+24t$; phase $-36t$; four copies each of $-90-6t$ minus and plus $3\sqrt{10}$ times $|24+t|$; and four copies each of $-171-6t$ minus and plus $3|24+t|$. For $-36 < t < -33$ all are positive and the energy $-144-4t$ lies below the origin. At $t=-34$ the energy is -8 . At $t=-34$, the spectrum is 3 with multiplicity four, $114-30\sqrt{10}$ with multiplicity four, 63 with multiplicity four, $114+30\sqrt{10}$ with multiplicity four, 624 and 1224. An independent eighteen-dimensional finite-difference Hessian reproduces this spectrum.

With the unique universal quark-source contraction this branch has one heavy family and two exactly massless light families before rank lifting. It supplies a stable hierarchy starting point without assigning three independent column weights. The complete joint potential, its rank-lifting operators and its CKM output have not been solved on this new field content. The kinetic theorem still applies if the completed branch has full-rank nondegenerate spectra and the required nonzero CP triangle.

What the result changes

A viable next UV construction must match its scalar potential, Yukawa contractions and matter metric together. The shared label group gives a justified contraction count and a heavy-family source branch, but internal symmetry with unrestricted allowed kinetic coefficients does not by itself protect an exact golden CKM equation at a CP-breaking vacuum. A finite-order prediction remains possible if omitted terms are controlled; an exact all-order relation requires stronger coefficient correlations. Merely adding phase charges, changing the group order or selecting a scalar stationary point does not supply those correlations for the operators constructed here.

The general importance of canonical normalization is established prior work. King and Peddie, Canonical normalisation and Yukawa matrices, Physics Letters B586 (2004) 83-94, arXiv:hep-ph/0312237, explicitly demonstrate changes in flavor textures. Espinosa and Ibarra, Flavour Symmetries and Kahler Operators, JHEP 0408 (2004) 010, arXiv:hep-ph/0405095, analyze its physical mixing effects and find subdominant quark effects under their phenomenological assumptions. Chen, Ramos-Sanchez and Ratz, A note on the predictions of models with modular flavor symmetries, arXiv:1909.06910, discuss additional Kahler parameters and predictivity. The contribution established in this repository is the explicit nine-polynomial spanning construction, its positive finite-degree completion, fixed-spectrum bare-mediator matching and reproducible application to the current joint vacuum. Publication priority has not been established by this targeted literature check.

Reproduction and certificates

Run `OPENBLAS_NUM_THREADS=1 PYTHONPATH=python python python/develop_flavor_kinetic.py` from the repository root. The inputs are the retained joint-vacuum and exact invariant-tensor receipts; the output is `receipts/m22_interactions/flavor_kinetic.json`. The module `python/perfectpower/flavor_kinetic.py` contains the reusable interpolating covariants, positive completion and bare matching. The focused tests cover exact symbolic identities, nine-dimensional spanning, family/phase/CP covariance away from the reference point, positive completion away from the vacuum, independent heavy-null-space matching, the retained four-direction response, the fixed-anchor golden deformation and all eighteen source masses. A fresh extraction repeats the scientific receipt byte for byte. The complete monograph and source archive retain the previous calculations and their stated boundaries.